Transit Hub Concentration & Passenger Density Topology

TOP 5 HUBS SHARE — 18.24% Total
2.73M
Cumulative volume across top 5 interchange hubs

TOP 2 HUBS SHARE — 8.86% Share
1.32M
Cairo University & Al-Shohadaa combined volume

AL-SHOHADAA VOLUME — Central Hub
658,870
4.40% network volume through Al-Shohadaa station

Operational Analysis: Relational aggregation across passenger flow records isolated acute passenger concentration within key transfer interchanges. The top 5 stations alone absorb 2.73M passengers (18.24% network share), led by Cairo University (667.4k, 4.46%) and Al-Shohadaa (658.9k, 4.40%) which channel 1.32M journeys (8.86%) through platforms that experience critical morning peak crowding.

01_station_density.sql — Execution Grid — SQL Server 2022

```
-- Top 5 Stations by Passenger Flow SELECT TOP 5 s.StationNameArabic, s.StationTypeArabic,
FORMAT(SUM(p.PassengerCount), 'N0') AS TotalPassengers, FORMAT(SUM(p.PassengerCount) * 1.0 / (SELECT
SUM(PassengerCount) FROM PassengerFlow), 'P2') AS PassengerPercentage FROM PassengerFlow p JOIN Stations s ON
p.StationID = s.StationID GROUP BY s.StationNameArabic, s.StationTypeArabic ORDER BY SUM(p.PassengerCount)
DESC;
```

SSMS OUTPUT GRID (TOP 5 STATIONS) — TIME: 00:00:00

#	StationNameArabic	StationTypeArabic	TotalPassengers	PassengerPercentage
1	جامعة القاهرة	تبادلية	667,410	4.46%
2	الشهداء	تبادلية	658,870	4.40%
3	السادات	تبادلية	520,366	3.48%
4	العتبة	تبادلية	514,110	3.43%
5	ناصر	تبادلية	369,625	2.47%

Query executed successfully — Rows: 5 | Aggregation: SUM, Subquery Ratio, Inner JOIN

Prepared by Mohamed Gomaa Mohamed | Data Analytics & BI
Dataset: Simulated data created for analytical and educational purposes; findings do not represent official Cairo Metro operational statistics.

Line Volume Distribution & Fleet Operational Load Analysis

LINE 1 VOLUME — Spine Corridor
40.89%
6,121,421 commuters carried across Line 1

LINE 3 EXPANSION — Modern Fleet
33.51%
5,016,964 passengers absorbed by Line 3

LINE 2 ABSORPTION — Cross-Capital
25.60%
3,832,575 passengers on Shobra-Giza line

Operational Analysis: Line 1 represents the primary transit corridor across the network, transporting more than 6.12M passengers (40.89% market share). With Line 3 absorbing 33.51% and Line 2 carrying 25.60%, fleet deployment and operational scheduling on Line 1 remain central to overall network throughput.

02_line_market_share.sql — Execution Grid — SQL Server 2022

```
-- Passenger Volume & Market Share per Line SELECT l.LineNameArabic, FORMAT(SUM(p.PassengerCount), 'N0') AS TotalPassengers, FORMAT(SUM(p.PassengerCount) * 1.0 / (SELECT SUM(PassengerCount) FROM PassengerFlow), 'P2') AS LineSharePercentage FROM PassengerFlow p JOIN Lines l ON p.LineID = l.LineID GROUP BY l.LineNameArabic ORDER BY SUM(p.PassengerCount) DESC;
```

SSMS OUTPUT GRID (LINES MARKET SHARE) — TIME: 00:00:00

#	LineNameArabic	TotalPassengers	LineSharePercentage
1	الخط الأول	6,121,421	40.89%
2	الخط الثالث	5,016,964	33.51%
3	الخط الثاني	3,832,575	25.60%

Query executed successfully — Rows: 3 | Technique: Relational Aggregation & Market Share Subquery

Morning Rush Bottleneck & CTE Pre-Aggregation Architecture

MORNING DELAY RATE — Peak Critical

89.37%

Trips delayed during morning rush hours

MORNING PASSENGERS — Surge Load

6,016,723

Commuters entering network during morning period

PEAK AVG LATENCY — +33% Vs Off-Peak

6.0 Mins

Average delay per trip (compared to 4.5m baseline)

Operational Analysis: Designed a Multi-CTE architecture to independently pre-aggregate passenger volumes and trip performance metrics before joining, eliminating many-to-many fan-out row multiplication. The results confirm morning demand surges to 6,016,723 passengers, coinciding with an 89.37% trip delay rate and a 6.0-minute average delay.

03_morning_rush_bottleneck.sql — CTE Grid — SQL Server 2022

```
WITH PassengerSummary AS ( SELECT TimePeriod, SUM(PassengerCount) AS TotalPassengers FROM
PassengerFlow GROUP BY TimePeriod ), TripSummary AS ( SELECT TimePeriod, COUNT(TripID) AS TotalTrips,
AVG(DelayMinutes * 1.0) AS AvgDelayMinutes, SUM(CASE WHEN DelayMinutes > 0 THEN 1.0 ELSE 0.0 END) /
COUNT(TripID) AS DelayedTripsRate FROM TrainTrips GROUP BY TimePeriod ) SELECT p.TimePeriod,
FORMAT(p.TotalPassengers, 'N0') AS TotalPassengers, FORMAT(t.AvgDelayMinutes, 'N1') AS AvgDelayMinutes,
FORMAT(t.DelayedTripsRate, 'P2') AS DelayedTripsRate FROM PassengerSummary p JOIN TripSummary t ON
p.TimePeriod = t.TimePeriod ORDER BY t.AvgDelayMinutes DESC;
```

SSMS OUTPUT GRID (PRE-AGGREGATED JOIN RESULTS) — TIME: 00:00:00

#	TimePeriod	TotalPassengers	AvgDelayMinutes	DelayedTripsRate
1	Morning	6,016,723	6.0	89.37%
2	Evening	5,295,421	4.5	84.08%
3	Afternoon	3,658,816	4.5	85.07%

Query executed successfully — Rows: 3 | Architecture: Dual-CTE Pre-Aggregation Join Engine

Incident Root Cause Categorization & Fleet Reliability

CROWD & DELAY SHARE — Root Cause
54.20%
271 of 500 total incidents caused by crowding & service delay

TECHNICAL FAULTS — Rolling Stock
20.40%
102 incidents attributed to mechanical/electrical issues

PASSENGER ASSIST — High Severity
15 High
Highest single severity tier (14.6% total share)

Operational Analysis: The analysis shows that crowding and service delays account for 54.2% of incidents in the simulated dataset (164 overcrowding, 107 service delays out of 500 total). Mechanical and electrical technical issues account for 20.4% (102 incidents), indicating that operational passenger flow management is a critical factor alongside fleet maintenance.

04_incident_root_causes.sql — Execution Grid — SQL Server 2022

```
-- Incident Root Causes SELECT IncidentType, COUNT(IncidentID) AS TotalIncidents,
FORMAT(COUNT(IncidentID) * 1.0 / (SELECT COUNT(*) FROM Incidents), 'P2') AS IncidentPercentage, SUM(CASE WHEN
Severity = 'High' THEN 1 ELSE 0 END) AS HighSeverityCount FROM Incidents GROUP BY IncidentType ORDER BY
TotalIncidents DESC;
```

SSMS OUTPUT GRID (INCIDENT TYPES BREAKDOWN) — TIME: 00:00:00

#	IncidentType	TotalIncidents	IncidentPercentage	HighSeverityCount
1	ازدحام	164	32.80%	13
2	تأخير خدمة	107	21.40%	14
3	مشكلة فنية	102	20.40%	10
4	مساعدة ركاب	73	14.60%	15
5	مشكلة مرافق	54	10.80%	13

Query executed successfully — Rows: 5 | Aggregation: Categorical Grouping, CASE Severity COUNT

Station Safety Risks & Non-Hub Incident Distribution

SAINT TERESA RISK — Peak Critical

5 High

Highest high-severity incident count among ranked stations

STANDARD STATION SHARE — Top 5 Cluster

5 of 5

All top incident stations are standard (non-transfer) stations

TOP CLUSTER INCIDENTS — Verified Sum

57 Events

Cumulative incidents across the 5 ranked standard stations

Operational Analysis: Evaluating incident distribution indicates notable variation across station types. The top 5 stations by incident count are all standard stations (57 total events across El-Haykestep, Saint Teresa, Dokki, Dar El-Salam, and Ain Shams), with Saint Teresa registering 5 high-severity cases. Further operational telemetry is required to isolate causal drivers.

05_station_safety_risks.sql — Execution Grid — SQL Server 2022

```
-- Station Safety Risks SELECT TOP 5 s.StationNameArabic, s.StationTypeArabic, COUNT(i.IncidentID) AS
TotalIncidents, SUM(CASE WHEN i.Severity = 'High' THEN 1 ELSE 0 END) AS HighSeverityIncidents FROM Incidents
i JOIN Stations s ON i.StationID = s.StationID GROUP BY s.StationNameArabic, s.StationTypeArabic ORDER BY
TotalIncidents DESC;
```

SSMS OUTPUT GRID (TOP VULNERABLE STATIONS) — TIME: 00:00:00

#	StationNameArabic	StationTypeArabic	TotalIncidents	HighSeverityIncidents
1	الهايكستب	عادية	12	2
2	سانت تريزا	عادية	12	5
3	الدقي	عادية	11	3
4	دار السلام	عادية	11	1
5	عين شمس	عادية	11	2

Query executed successfully — Rows: 5 | Filter: TOP 5 Vulnerable Stations by Incident Frequency

Hourly Delay Cascade Analysis via Window Function LAG()

PEAK SHOCK HOUR — 08:00 Peak
6.21 Mins
Highest hourly delay average across 177 trips

MORNING SHOCK DELTA — +0.39 Mins
+0.39m
Hour-over-hour delay surge via LAG() from 07:00

EVENING COMPARISON — 18:00 Peak
4.76 Mins
1.45m lower than morning peak average (6.21m)

Operational Analysis: Implementing LAG() OVER (ORDER BY OperationHour) tracks hour-over-hour latency shifts. The analysis identified an increase in average delay from 5.82 minutes at 07:00 to 6.21 minutes at 08:00, followed by a decrease to 6.03 minutes at 09:00. Evening delays peaked at 4.76 minutes at 18:00, remaining 1.45 minutes below the morning peak.

06_hourly_delay_cascade.sql — Window Output — SQL Server 2022

```
WITH HourlyMetrics AS ( SELECT DATEPART(HOUR, DepartureTime) AS OperationHour, COUNT(TripID) AS TripsCount, ROUND(AVG(DelayMinutes * 1.0), 2) AS CurrentHourAvgDelay FROM TrainTrips GROUP BY DATEPART(HOUR, DepartureTime) ) SELECT OperationHour, TripsCount, CurrentHourAvgDelay, LAG(CurrentHourAvgDelay) OVER (ORDER BY OperationHour) AS PreviousHourAvgDelay, ROUND(CurrentHourAvgDelay - LAG(CurrentHourAvgDelay) OVER (ORDER BY OperationHour), 2) AS DelayChange FROM HourlyMetrics ORDER BY OperationHour;
```

SSMS OUTPUT GRID (TEMPORAL CASCADE WINDOW) — TIME: 00:00:00

#	Hour	Trips	CurrentAvgDelay	PrevAvgDelay	DelayChange
1	07:00	180	5.820000	NULL	NULL
2	08:00	177	**6.210000**	5.820000	**+0.390000**
3	09:00	179	6.030000	6.210000	-0.180000
4	14:00	180	4.770000	6.030000	-1.260000
5	15:00	179	4.200000	4.770000	-0.570000
6	16:00	177	4.450000	4.200000	+0.250000
7	18:00	177	4.760000	4.450000	+0.310000

Query executed successfully — Rows: 3 | Technique: Relational Aggregation & Market Share Subquery

Prepared by Mohamed Gomaa Mohamed | Data Analytics & Business Intelligence
Dataset: Simulated data created for analytical and educational purposes; findings do not represent official Cairo Metro operational statistics.

● Query executed successfully
Rows: 5 | Aggregation: Categorical Grouping, CASE Severity COUNT

Query executed successfully

Rows: 5 | Filter: TOP 5 Vulnerable Stations by Incident Frequency

Dataset: Simulated data created for analytical and educational purposes; findings do not represent official Cairo Metro operational statistics.

Query executed successfully

Rows: 9 | Technique: CTE + LAG() Analytical Window Function

Dataset: Simulated data created for analytical and educational purposes; findings do not represent official Cairo Metro operational statistics.